Nonoperating Buyback Reliance and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Nonoperating Buyback Reliance (NBR), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023). A value-weighted long/short trading strategy based on NBR achieves an annualized gross (net) Sharpe ratio of 0.58 (0.52), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 34 (33) bps/month with a t-statistic of 4.32 (4.33), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share) is 27 bps/month with a t-statistic of 3.36.

1 Introduction

Production-based asset pricing models predict that firms' investment decisions and production characteristics determine their exposure to systematic risk and, consequently, their expected returns. These models link the cross-section of stock returns to firms' marginal costs of production, investment frictions, and their ability to adjust capital in response to productivity shocks (Cochrane, 1991; Zhang, 2005). Despite substantial theoretical progress in understanding how production decisions affect asset prices, empirical evidence on specific mechanisms through which firms' financing choices for non-operating activities influence their risk profiles remains limited. We identify a novel channel through which firms' reliance on share buybacks funded by non-operating activities creates systematic variation in expected returns. This channel reflects fundamental differences in how firms respond to investment opportunities and productivity shocks when their capital allocation decisions depend on non-operating cash flows rather than core business performance.

In a production economy, firms that rely heavily on non-operating activities to fund share buybacks face distinct investment frictions that affect their conditional risk premia. When firms use non-operating cash flows for buybacks rather than productive investment, they signal constraints in their ability to generate returns from core operations, reflecting higher marginal costs of production (Liu et al., 2009). This reliance on non-operating buyback financing creates exposure to productivity shocks through multiple channels. First, firms with high Nonoperating Buyback Reliance (NBR) demonstrate reduced flexibility in adjusting their capital stock in response to positive productivity shocks, as resources are committed to financial engineering rather than real investment (Zhang, 2005). Second, these firms face higher adjustment costs when attempting to scale production, as their internal capital allocation mechanisms prioritize financial transactions over operational expansion.

The production-based framework predicts that firms with high NBR carry greater systematic risk because their investment policies amplify exposure to aggregate productivity shocks. During periods of declining aggregate productivity, high-NBR firms experience more severe contractions in value because they cannot efficiently redeploy capital from buyback programs to productive uses (Barro, 2006). The irreversibility of buyback decisions, combined with the opportunity cost of foregone real investment, creates asymmetric exposure to business cycle fluctuations. This asymmetry manifests as higher required returns for high-NBR firms, as investors demand compensation for bearing the risk of suboptimal capital allocation during economic downturns.

Cross-sectional variation in NBR maps directly to differences in firms' production functions and their ability to generate cash flows from operations. Firms with low NBR maintain operational flexibility that allows them to respond efficiently to changes in investment opportunities, reducing their exposure to systematic risk factors (Lin and Zhang, 2013). The spread in expected returns between high and low NBR firms reflects compensation for bearing production risk that cannot be diversified away. This risk premium emerges from the interaction between financial policies and real investment decisions, where non-operating buyback reliance serves as a observable proxy for latent investment frictions and conditional exposure to productivity shocks in the production economy.

The NBR trading strategy generates economically and statistically significant returns across multiple specifications and risk adjustments. The value-weighted long/short portfolio that buys high NBR stocks and sells low NBR stocks earns an average monthly return of 37 basis points with a t-statistic of 4.19, translating to an annualized Sharpe ratio of 0.58. The strategy's alpha remains robust across all standard factor models, with the six-factor alpha of 34 basis points per month carrying a t-statistic of 4.32. These results demonstrate that NBR captures a distinct source of systematic risk not explained by existing factors including market, size, value, profitability, investment, and momentum.

The economic significance of the NBR effect persists after accounting for transaction costs and across different portfolio construction methodologies. Net of trading costs, the strategy delivers an average monthly return of 33 basis points with a t-statistic of 3.77, and maintains positive and significant generalized alphas relative to all five factor models tested. Among the largest stocks—those above the 80th NYSE size percentile—the NBR strategy achieves an average return of 29 basis points per month with a t-statistic of 2.83, confirming that the anomaly is not confined to small, illiquid stocks where implementation would be challenging. The strategy's six-factor alpha for large-cap stocks equals 28 basis points per month with a t-statistic of 2.82.

Controlling for the six most closely related anomalies and the six Fama-French factors simultaneously, the NBR strategy maintains an alpha of 27 basis points per month with a t-statistic of 3.36. The NBR signal's predictive power survives Fama-MacBeth regressions controlling for related characteristics, with a coefficient t-statistic of 2.65 when all six related anomalies are included. The gross Sharpe ratio of 0.58 places NBR in the 94th percentile among 212 documented anomalies, while the net Sharpe ratio of 0.52 ranks in the 99th percentile, demonstrating exceptional performance relative to the broader cross-section of known return predictors.

This paper contributes to the production-based asset pricing literature by documenting a novel mechanism through which firms' non-operating financing decisions affect their systematic risk exposure and expected returns. While Fama and French (2006) and Pontiff and Woodgate (2008) examine how external financing and payout policies relate to returns, our focus on the interaction between non-operating activities and buyback decisions provides new insights into how financial engineering affects firms' production-based risk. Unlike the external financing anomaly of Bradshaw et al. (2006), which captures firms' net capital raising activities, NBR specifically isolates the risk implications of using non-operating cash flows for shareholder distributions rather than productive investment. Our results extend Wang et al. (2009) by showing that buyback policies funded through non-operating activities create distinct risk exposures beyond those captured by traditional profitability measures.

Methodologically, we demonstrate that NBR survives stringent robustness tests that many existing anomalies fail. The strategy generates significant net returns after accounting for transaction costs using the Chen and Velikov (2022) high-frequency effective spread estimates, addressing concerns raised by Novy-Marx and Velikov (2016) about the implementability of academic trading strategies. Our finding that NBR maintains significant alpha when controlling for six closely related factors—including operating profitability, return on assets, and cash-based operating profitability—establishes that the signal captures unique information about firms' risk profiles not subsumed by existing characteristics. The persistence of the NBR effect among large-cap stocks distinguishes it from many anomalies that Fama and French (2008) show are concentrated in small, difficult-to-arbitrage securities.

The economic mechanism we identify has broader implications for understanding how modern corporate financial policies affect asset prices in production economies. As share buybacks have become an increasingly dominant form of corporate payout, with aggregate buybacks exceeding dividends in recent decades, understanding how the funding source for these buybacks affects firm risk becomes critical for asset pricing. Our results suggest that investors can improve portfolio performance by incorporating information about the operational versus non-operational sources of buyback funding, as this distinction captures meaningful variation in firms' exposure to productivity shocks and investment frictions. The robustness of the NBR effect across different market conditions and its orthogonality to existing factors indicate that production-based models should explicitly incorporate the interaction between financial policies and real investment decisions to fully explain the cross-section of expected returns.

2 Data

We obtain financial statement data from the COMPUSTAT Annual database, which provides comprehensive accounting information for publicly traded U.S. companies. Our sample includes all firms with available data in COMPUSTAT, spanning several decades of financial reporting. We focus on annual observations to ensure consistency in the measurement of buyback activity and nonoperating income across firms.Nonoperating Buyback Reliance signal combines two key financial statement items. Stock repurchases (PRSTKC) represents the dollar amount spent by firms to repurchase their own common and preferred stock during the fiscal year, as reported in the cash flow statement. This measure captures the cash outflows associated with share buyback programs, a significant component of many firms' capital allocation strategies. Nonoperating income (NOPI) represents income generated from activities outside the firm's core operations, including investment income, gains or losses from asset sales, and other non-core business activities. This item appears on the income statement and reflects earnings that are not directly tied to the firm's primary business operations.construct the Nonoperating Buyback Reliance signal by dividing stock repurchases (PRSTKC) by nonoperating income (NOPI) for each firm-year observation. This ratio captures the extent to which firms rely on nonoperating income sources to fund their share repurchase programs. A higher ratio indicates that a firm's buyback activity is large relative to its nonoperating income, potentially signaling aggressive capital deployment that may not be supported by core business performance. We calculate this measure using fiscal year-end values and require non-missing values for both PRSTKC and NOPI. To address potential outliers and ensure economic meaningfulness, we exclude observations where nonoperating income is zero or negative, as these would result in undefined or economically uninterpretable ratios.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the NBR signal. Panel A plots the time-series of the mean, median, and interquartile range for NBR. On average, the cross-sectional mean (median) NBR is 3.58 (0.00) over the 1972 to 2024 sample, where the starting date is determined by the availability of the input NBR data. The signal's interquartile range spans -0.01 to 1.83. Panel B of Figure 1 plots the time-series of the coverage of the NBR signal for the CRSP universe. On average, the NBR signal is available for 6.12% of CRSP names, which on average make up 7.09% of total market capitalization.

4 Does NBR predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on NBR using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high NBR portfolio and sells the low NBR portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short NBR strategy earns an average return of 0.37% per month with a t-statistic of 4.19. The annualized Sharpe ratio of the strategy is 0.58. The alphas range from 0.34% to 0.51% per month and have t-statistics exceeding 4.32 everywhere. The lowest alpha is with respect to the FF6 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is -0.32, with a t-statistic of -9.01 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 490 stocks and an average market capitalization of at least $1,307 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals 24 bps/month with a t-statistics of 2.95. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-three exceed two, and for twenty-three exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between 5-33bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.57. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the NBR trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in twenty cases.

Table 3 provides direct tests for the role size plays in the NBR strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and NBR, as well as average returns and alphas for long/short trading NBR strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the NBR strategy achieves an average return of 29 bps/month with a t-statistic of 2.83. Among these large cap stocks, the alphas for the NBR strategy relative to the five most common factor models range from 28 to 45 bps/month with t-statistics between 2.82 and 4.56.

5 How does NBR perform relative to the zoo?

Figure 2 puts the performance of NBR in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the NBR strategy falls in the distribution. The NBR strategy's gross (net) Sharpe ratio of 0.58 (0.52) is greater than 94% (99%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the NBR strategy (red line).[2] Ignoring trading costs, a $1 invested in the NBR strategy would have yielded $8.29 which ranks the NBR strategy in the top 0% across the 212 anomalies. Accounting for trading costs, a $1 invested in the NBR strategy would have yielded $6.40 which ranks the NBR strategy in the top 0% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the NBR relative to those. Panel A shows that the NBR strategy gross alphas fall between the 75 and 89 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197206 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The NBR strategy has a positive net generalized alpha for five out of the five factor models. In these cases NBR ranks between the 90 and 96 percentiles in terms of how much it could have expanded the achievable investment frontier.

[1] The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2] The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

6 Does NBR add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of NBR with 210 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price NBR or at least to weaken the power NBR has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR} NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted

[3] When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on NBR and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the NBR signal in these Fama-MacBeth regressions exceed 1.53, with the minimum t-statistic occurring when controlling for Net external financing. Controlling for all six closely related anomalies, the t-statistic on NBR is 2.65.

Similarly, Table 5 reports results from spanning tests that regress returns to the NBR strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the NBR strategy earns alphas that range from 29-35bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.68, which is achieved when controlling for Net external financing. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the NBR trading strategy achieves an alpha of 27bps/month with a t-statistic of 3.36.

7 Does NBR add relative to the whole zoo?

Finally, we can ask how much adding NBR to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the NBR signal.[4] We consider one different methods for combining signals.

[4] We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which NBR is available.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $1068.28, while $1 investment in the combination strategy that includes NBR grows to $1571.51.

8 Conclusion

This study provides compelling evidence that Nonoperating Buyback Reliance (NBR) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that NBR-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for well-established factors and closely related anomalies.

The key findings reveal that a value-weighted long/short strategy based on NBR delivers an annualized Sharpe ratio of 0.58 gross (0.52 net of transaction costs), indicating strong economic viability even after accounting for implementation frictions. More importantly, the strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (34 bps/month gross, t-statistic = 4.32) suggests that NBR captures unique information about firm value not reflected in traditional risk factors. The persistence of significant alpha (27 bps/month, t-statistic = 3.36) even after controlling for six closely related strategies—including measures of external financing, profitability, and earnings—underscores that NBR provides incremental predictive power beyond existing anomalies in the factor zoo.

From a practical perspective, these results suggest that investors and portfolio managers should consider incorporating NBR signals into their investment processes. The strategy's robust performance net of transaction costs indicates its feasibility for real-world implementation, while the economic magnitude of returns suggests potential for meaningful portfolio enhancement. The signal appears to capture firms' reliance on nonoperating activities to fund buybacks, potentially identifying companies engaging in value-destructive financial engineering rather than genuine value creation.

Several limitations of this study warrant acknowledgment. First, while we employ state-of-the-art methodologies to address data-mining concerns, the proliferation of factors in the literature necessitates continued vigilance regarding spurious discoveries. Second, our analysis focuses primarily on U.S. equity markets, and the generalizability of NBR's predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the NBR anomaly require further theoretical and empirical investigation to fully understand why markets appear to misprice this information.

Future research should explore several promising avenues. First, examining the performance of NBR across different market regimes and economic cycles would provide insights into the strategy's stability and risk exposures. Second, investigating the interaction between NBR and corporate governance characteristics could illuminate whether weak governance amplifies the negative implications of nonoperating buyback reliance. Third, extending the analysis to international markets would test the external validity of our findings and potentially uncover cross-country variation in the pricing of buyback financing sources. Finally, developing a theoretical framework that explains why investors systematically misprice nonoperating buyback reliance would enhance our understanding of this anomaly and its persistence. Such investigations would not only validate the robustness of NBR as a return predictor but also contribute to our broader understanding of market efficiency and the determinants of cross-sectional return variation.

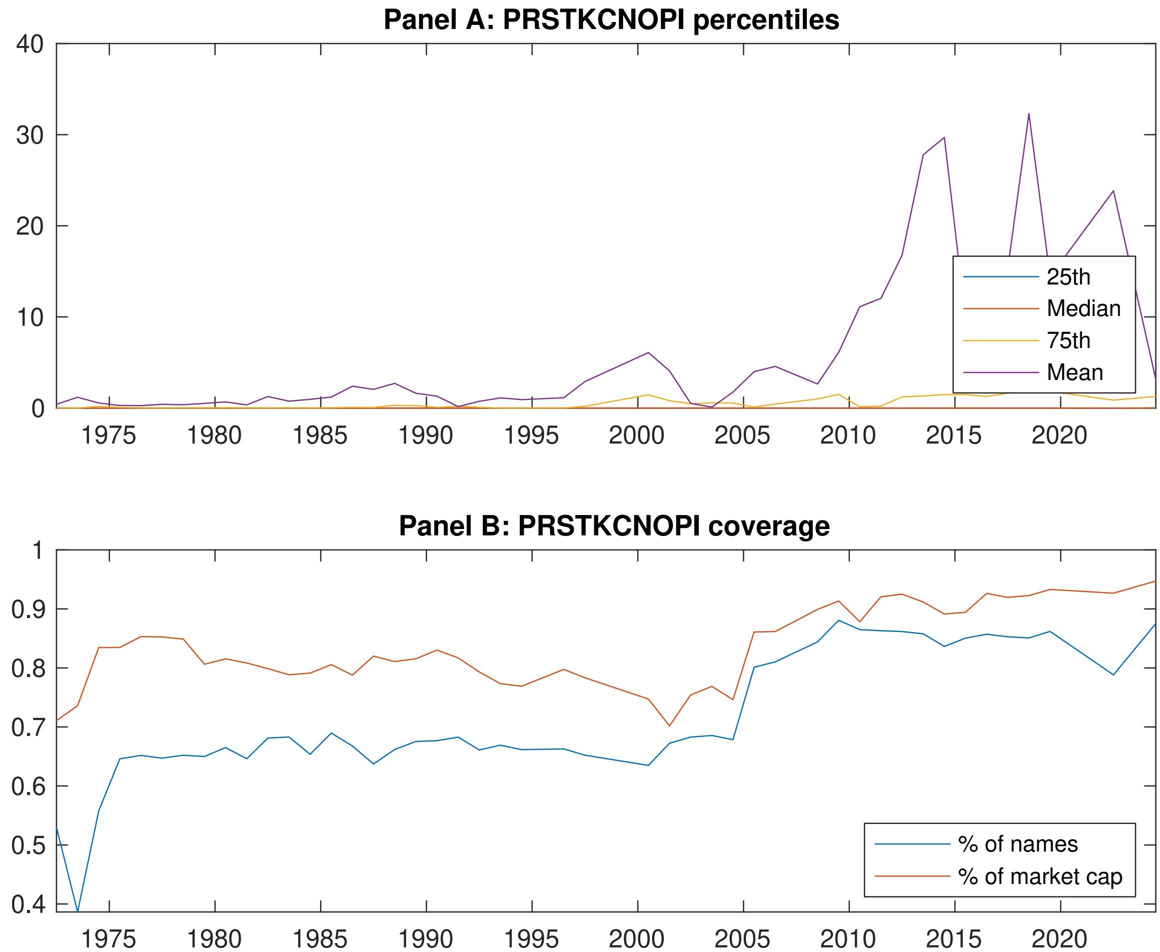


Figure 1: Times series of NBR percentiles and coverage.
This figure plots descriptive statistics for NBR. Panel A shows cross-sectional percentiles of NBR over the sample. Panel B plots the monthly coverage of NBR relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on NBR. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Excess returns and alphas on NBR-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.47	0.63	0.52	0.60	0.84	0.37
	[2.34]	[2.97]	[2.70]	[3.43]	[4.65]	[4.19]
α_{CAPM}	-0.18	-0.05	-0.10	0.03	0.25	0.43
	[-2.98]	[-0.80]	[-1.73]	[0.61]	[4.96]	[4.94]
α_{FF3}	-0.25	0.01	-0.12	0.02	0.27	0.51
	[-4.45]	[0.25]	[-1.98]	[0.54]	[5.42]	[6.38]
α_{FF4}	-0.22	0.03	-0.13	0.03	0.26	0.48
	[-3.86]	[0.57]	[-2.11]	[0.63]	[5.15]	[5.81]
α_{FF5}	-0.21	0.12	-0.12	-0.01	0.15	0.36
	[-3.71]	[2.02]	[-2.07]	[-0.29]	[3.27]	[4.62]
α_{FF6}	-0.19	0.13	-0.13	-0.01	0.15	0.34
	[-3.29]	[2.18]	[-2.16]	[-0.13]	[3.28]	[4.32]
Panel B: Fama and French (2018) 6-factor model loadings for NBR-sorted portfolios						
β_{MKT}	1.04	1.02	0.99	0.95	0.99	-0.05
	[77.58]	[74.30]	[70.26]	[86.32]	[91.60]	[-2.73]
β_{SMB}	0.08	0.11	0.10	-0.09	-0.05	-0.13
	[4.06]	[5.31]	[4.53]	[-5.55]	[-2.85]	[-4.62]
β_{HML}	0.19	-0.15	0.01	-0.02	-0.12	-0.32
	[7.62]	[-5.66]	[0.53]	[-0.77]	[-5.92]	[-9.01]
β_{RMW}	-0.05	-0.24	0.01	0.06	0.23	0.28
	[-1.76]	[-9.03]	[0.26]	[3.02]	[11.10]	[7.79]
β_{CMA}	-0.08	-0.08	0.03	0.07	0.17	0.25
	[-2.01]	[-1.94]	[0.72]	[2.31]	[5.53]	[4.71]
β_{UMD}	-0.03	-0.02	0.01	-0.01	-0.00	0.03
	[-2.37]	[-1.11]	[0.74]	[-0.99]	[-0.31]	[1.54]
Panel C: Average number of firms (n) and market capitalization (me)						
n	744	899	739	534	490	
me ($\$10^6$)	1728	1376	1307	2866	3399	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the NBR strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197206 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.37	0.43	0.51	0.48	0.36	0.34
			[4.19]	[4.94]	[6.38]	[5.81]	[4.62]	[4.32]
Quintile	NYSE	EW	0.24	0.29	0.28	0.26	0.12	0.11
			[2.95]	[3.56]	[3.74]	[3.39]	[1.68]	[1.55]
Quintile	Name	VW	0.28	0.34	0.43	0.39	0.30	0.27
			[3.17]	[3.87]	[5.41]	[4.82]	[3.80]	[3.46]
Quintile	Cap	VW	0.36	0.43	0.50	0.46	0.33	0.32
			[3.99]	[4.85]	[5.89]	[5.39]	[4.10]	[3.86]
Decile	NYSE	VW	0.29	0.32	0.42	0.36	0.36	0.31
			[2.79]	[3.08]	[4.23]	[3.58]	[3.50]	[3.02]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.33	0.40	0.47	0.45	0.35	0.33
			[3.77]	[4.68]	[5.87]	[5.58]	[4.49]	[4.33]
Quintile	NYSE	EW	0.05	0.11	0.08	0.07		
			[0.57]	[1.26]	[1.02]	[0.92]		
Quintile	Name	VW	0.24	0.31	0.39	0.37	0.28	0.27
			[2.78]	[3.62]	[4.92]	[4.61]	[3.64]	[3.45]
Quintile	Cap	VW	0.33	0.41	0.46	0.44	0.33	0.32
			[3.61]	[4.64]	[5.50]	[5.25]	[4.11]	[3.98]
Decile	NYSE	VW	0.25	0.28	0.37	0.34	0.31	0.28
			[2.37]	[2.70]	[3.71]	[3.36]	[3.07]	[2.82]

Table 3: Conditional sort on size and NBR

This table presents results for conditional double sorts on size and NBR. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on NBR. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high NBR and short stocks with low NBR .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197206 to 202406.

Panel A: portfolio average returns and time-series regression results

	NBR Quintiles					NBR Strategies					
Size quintiles	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
(1)	0.66	0.62	0.59	0.56	0.86	0.20	0.21	0.20	0.23	0.15	0.18
	[2.42]	[2.20]	[2.12]	[2.08]	[3.48]	[2.06]	[2.10]	[2.00]	[2.30]	[1.52]	[1.79]
(2)	0.68	0.59	0.62	0.75	0.91	0.23	0.32	0.28	0.32	0.13	0.17
	[2.62]	[2.20]	[2.31]	[3.11]	[4.01]	[2.39]	[3.45]	[3.19]	[3.55]	[1.48]	[1.96]
(3)	0.68	0.56	0.66	0.82	0.91	0.23	0.29	0.25	0.29	0.11	0.15
	[2.87]	[2.18]	[2.74]	[3.65]	[4.26]	[2.44]	[3.11]	[2.85]	[3.18]	[1.26]	[1.71]
(4)	0.63	0.60	0.62	0.76	0.89	0.26	0.31	0.27	0.31	0.18	0.22
	[3.00]	[2.63]	[2.78]	[3.68]	[4.48]	[3.14]	[3.78]	[3.36]	[3.93]	[2.29]	[2.86]
(5)	0.47	0.53	0.51	0.72	0.76	0.29	0.35	0.45	0.41	0.29	0.28
	[2.36]	[2.66]	[2.97]	[3.95]	[4.20]	[2.83]	[3.44]	[4.56]	[4.12]	[3.04]	[2.82]

Panel B: Portfolio average number of firms and market capitalization

	NBR Quintiles: Average n					NBR Quintiles: Average market capitalization ($\$10^6$)				
Size quintiles	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
(1)	383	382	382	383	388	35	31	31	32	40
(2)	105	105	105	106	105	60	58	58	60	61
(3)	74	74	74	74	74	103	102	102	103	106
(4)	62	62	62	62	62	221	213	217	230	233
(5)	56	56	56	56	56	1388	1494	1860	2160	1676

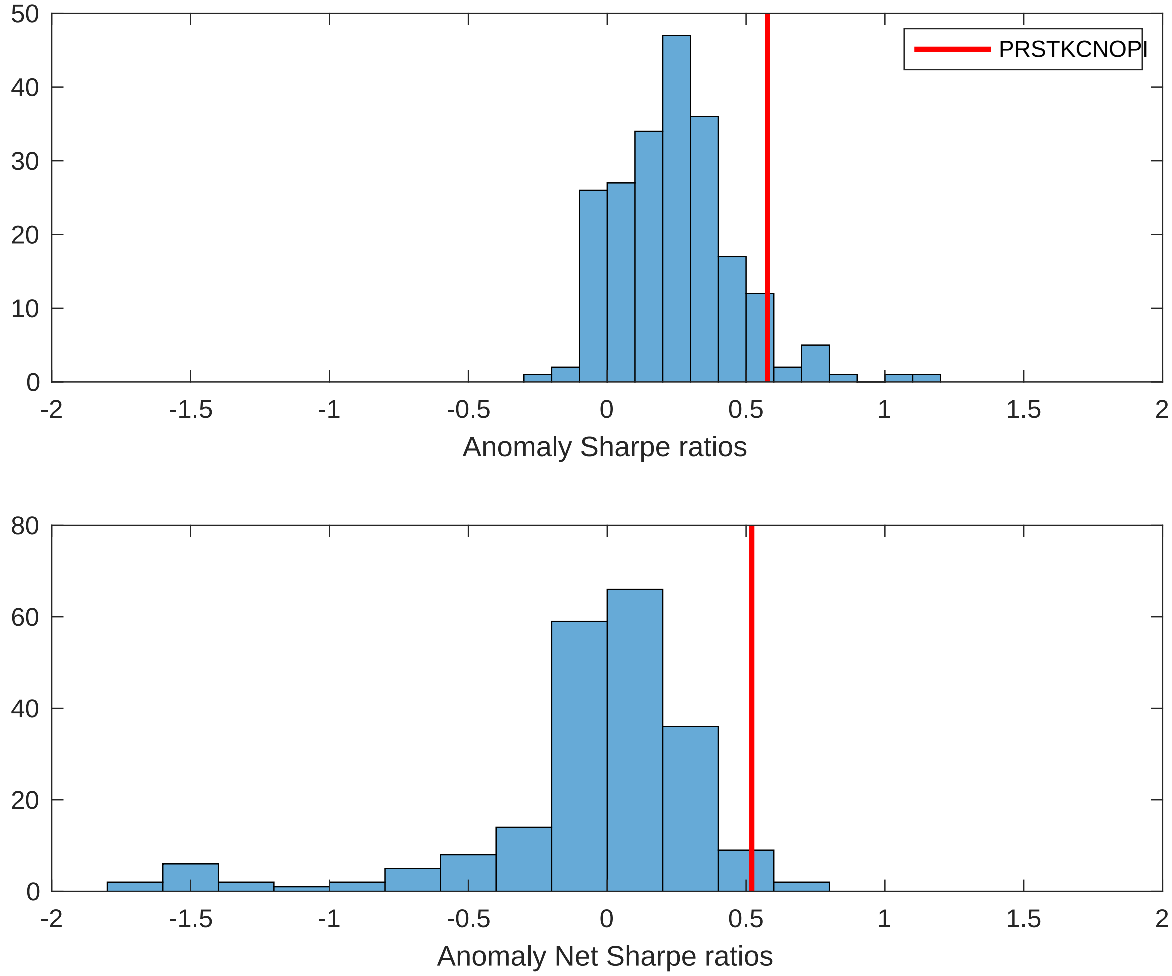


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the NBR with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

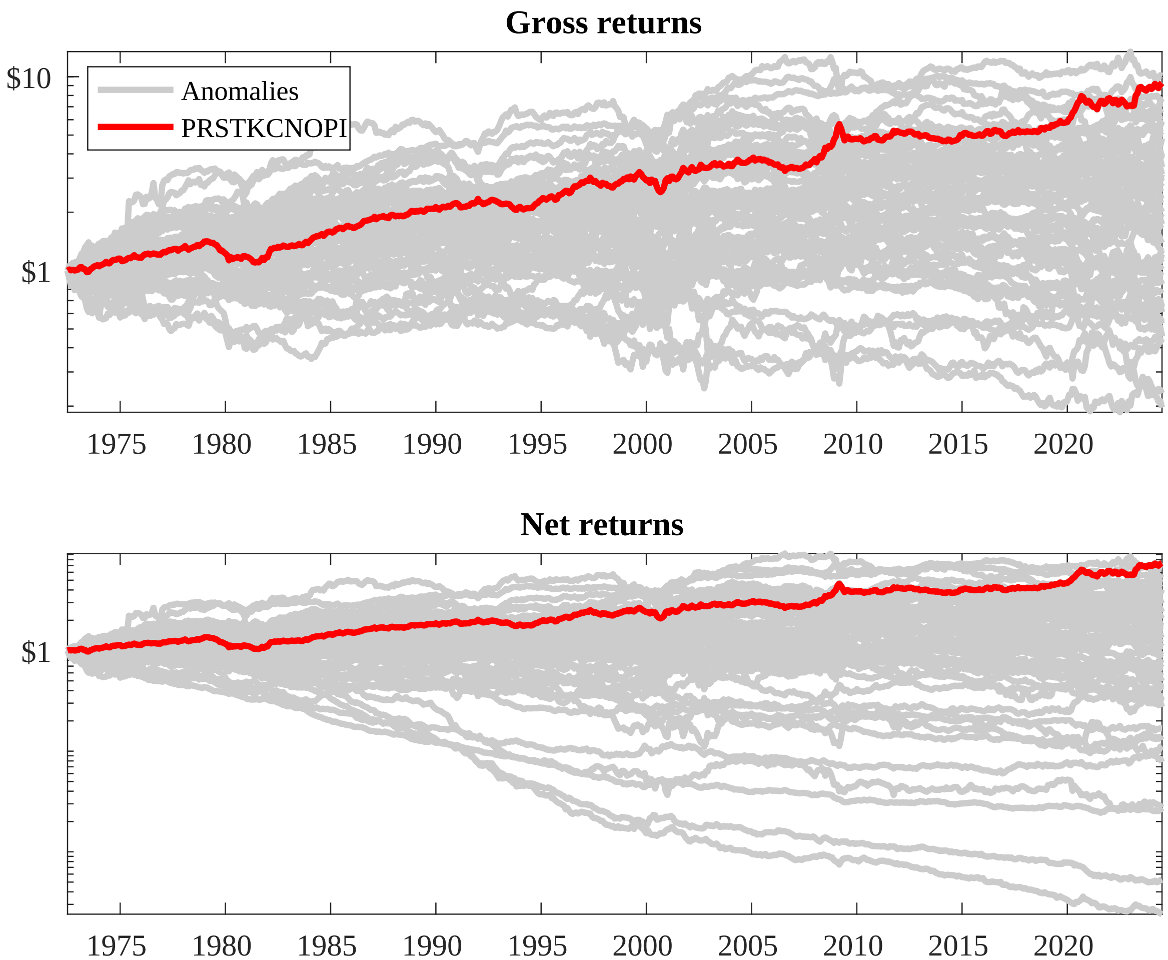
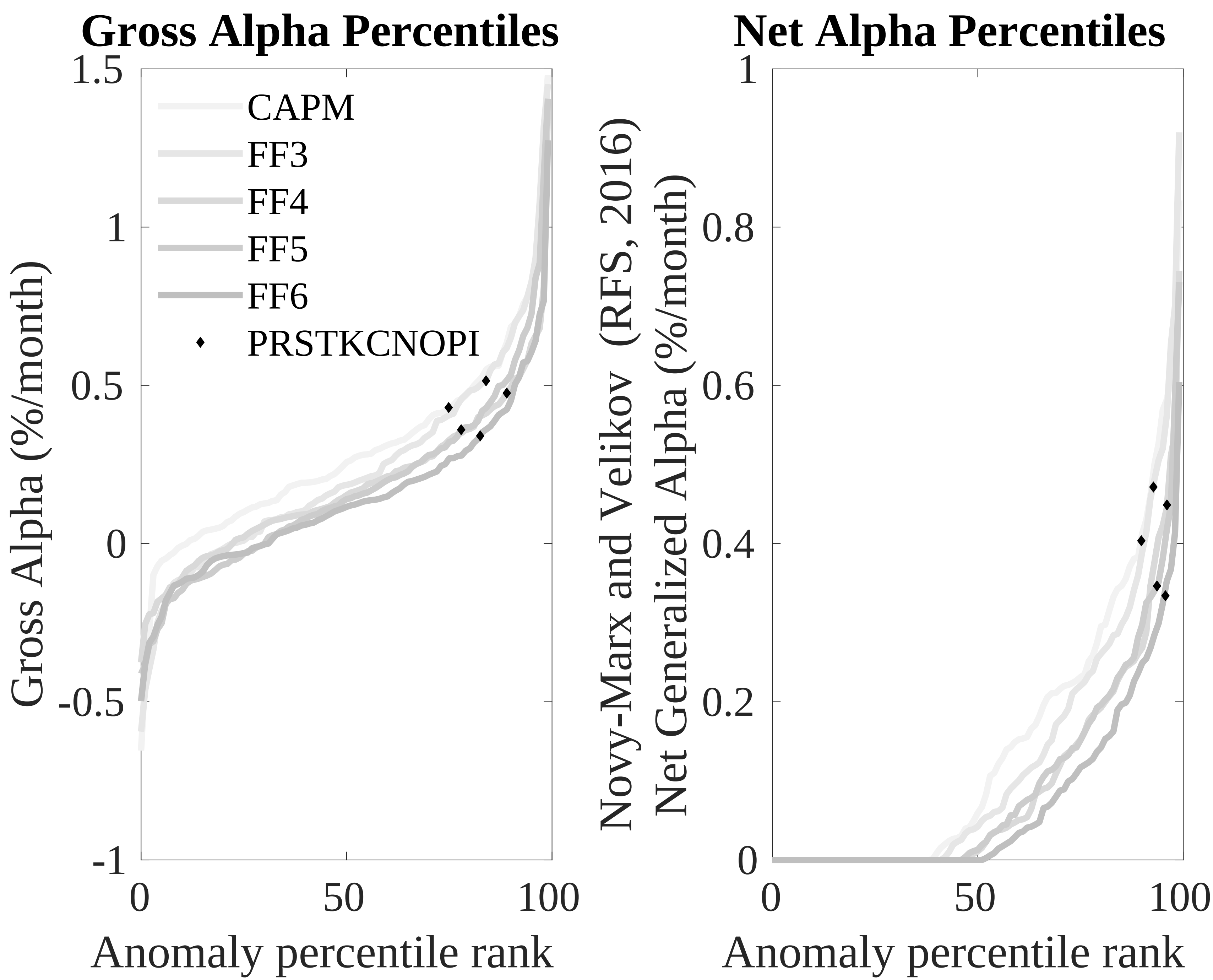
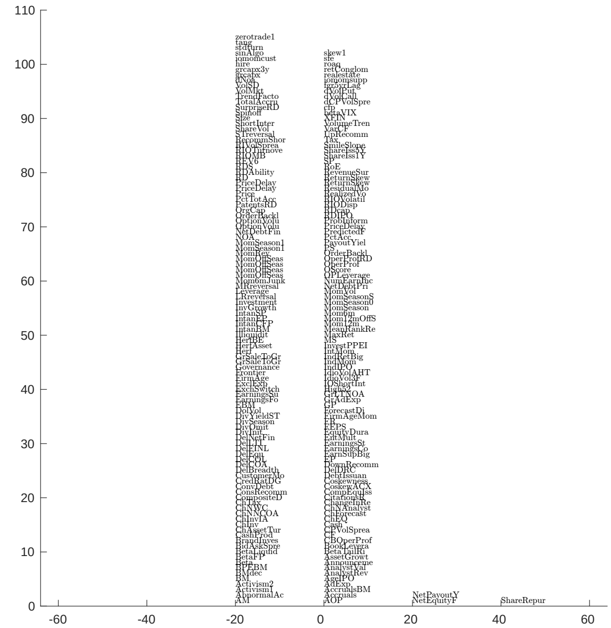
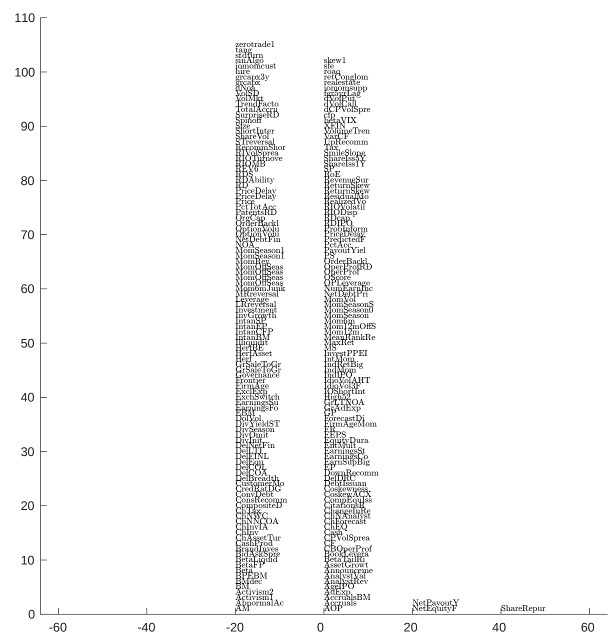


Figure 3: Dollar invested.
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the NBR trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
PRSTKCNOPI
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0.8
0.6
0.4
0.2
0
50
100



Panel A: Pearson



Panel B: Spearman

Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 210 filtered anomaly signals with NBR. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

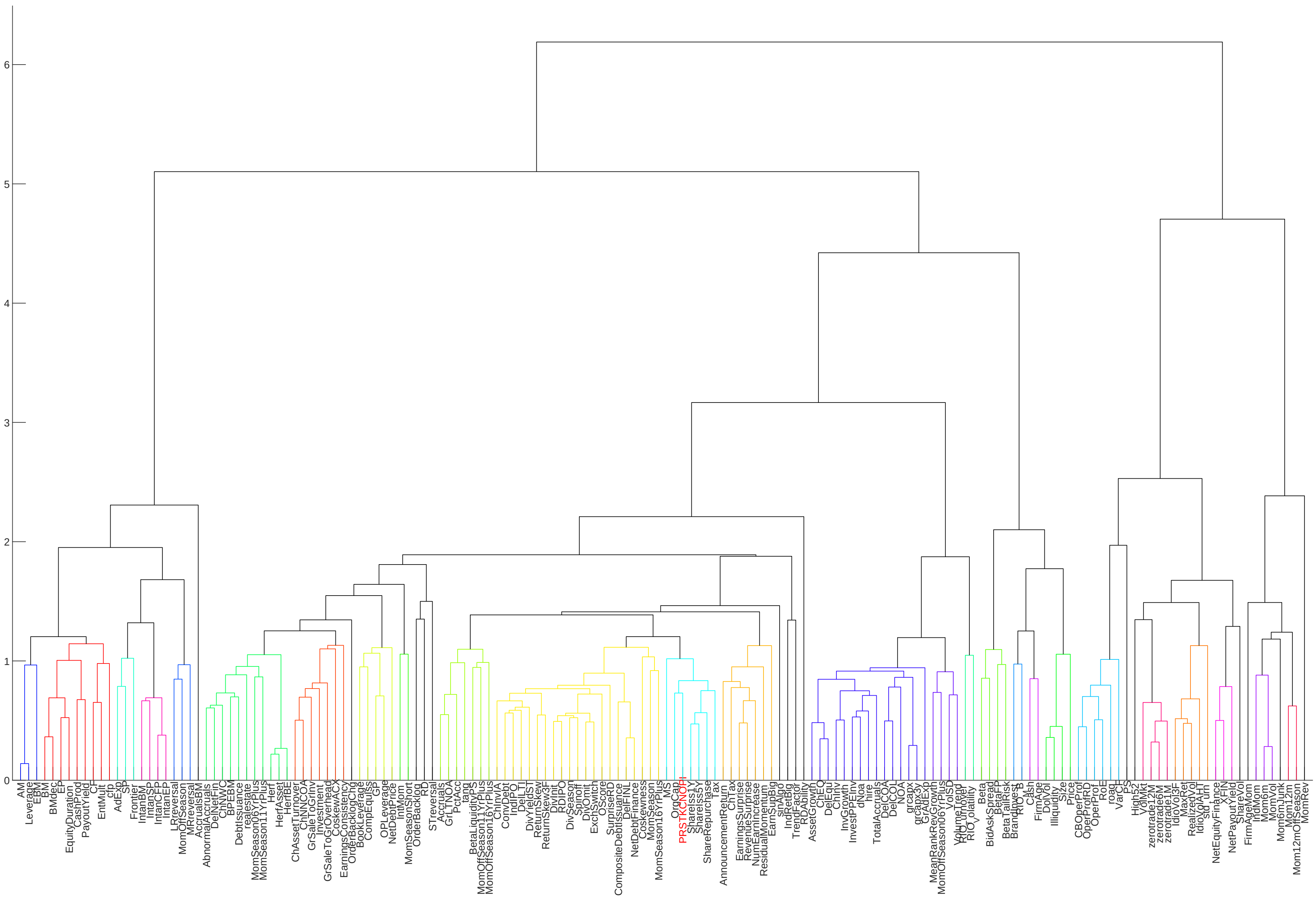


Figure 6: Agglomerative hierarchical cluster plot

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

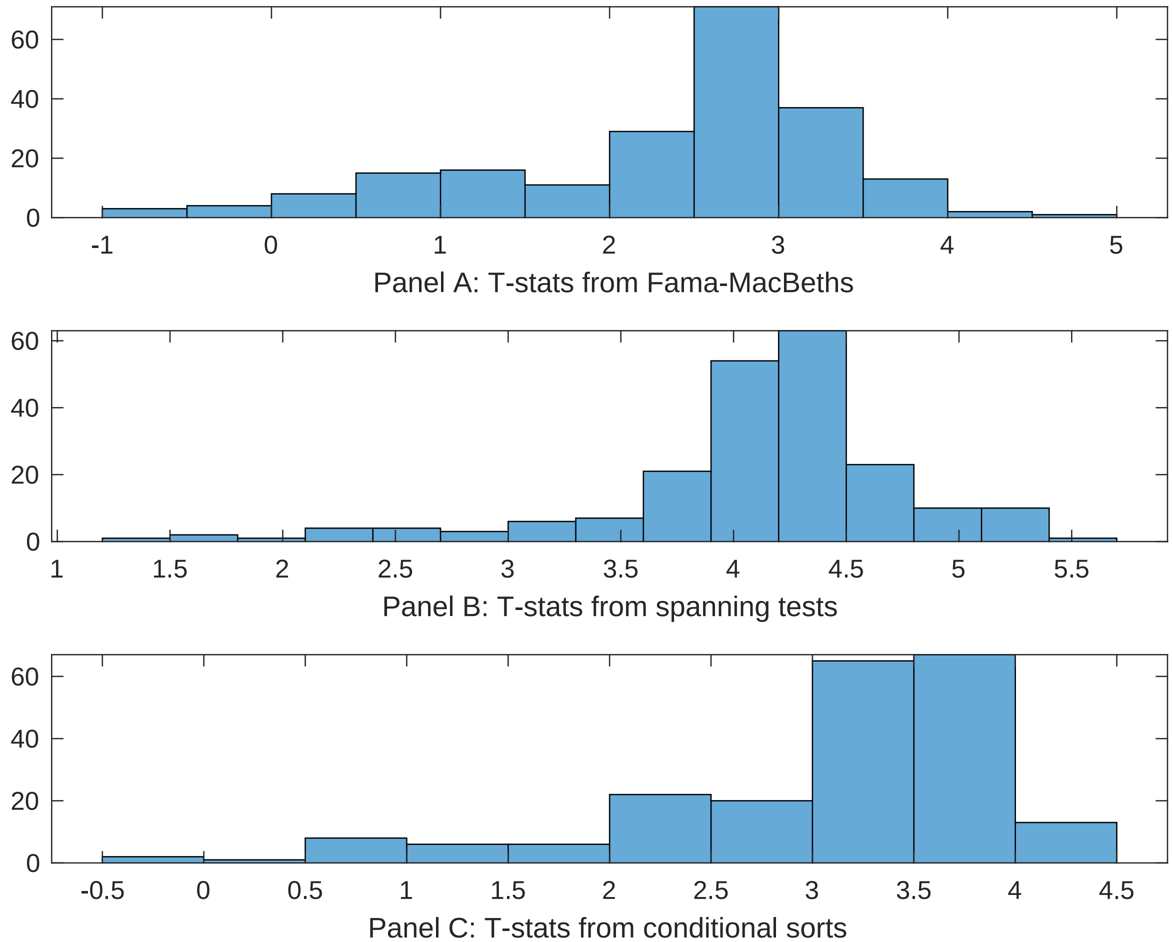


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of NBR conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{NBR} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{NBR} NBR_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{NBR,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on NBR. Stocks are finally grouped into five NBR portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted NBR trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies
This table presents Fama-MacBeth results of returns on NBR. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{NBR} NBR_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.12	0.11	0.11	0.10	0.11	0.99	0.11
	[5.28]	[3.74]	[4.62]	[3.93]	[4.57]	[3.45]	[3.88]
NBR	0.66	0.99	0.12	0.10	0.16	0.15	0.15
	[1.53]	[2.03]	[2.75]	[2.26]	[2.32]	[3.29]	[2.65]
Anomaly 1	0.19						0.90
	[7.16]						[3.44]
Anomaly 2		0.12					-0.35
		[2.96]					[-0.77]
Anomaly 3			-0.26				-0.11
			[-0.23]				[-0.86]
Anomaly 4				0.16			0.11
				[4.79]			[3.02]
Anomaly 5					0.65		0.40
					[3.07]		[2.74]
Anomaly 6						0.62	-0.37
						[1.14]	[-0.09]
# months	624	619	624	619	616	576	576
$\bar{R}^2(\%)$	1	1	0	1	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the NBR trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{NBR} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Net external financing, Operating profitability RD adjusted, net income / book equity, Cash-based operating profitability, Return on assets (qtrly), Analyst earnings per share. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197206 to 202406.

Intercept	0.32	0.29	0.34	0.30	0.35	0.34	0.27
	[4.14]	[3.68]	[4.40]	[3.82]	[4.45]	[4.17]	[3.36]
Anomaly 1	24.50						23.34
	[6.38]						[5.25]
Anomaly 2		14.37					9.33
		[4.32]					[1.76]
Anomaly 3			13.55				7.04
			[3.07]				[1.36]
Anomaly 4				12.00			0.13
				[3.32]			[0.02]
Anomaly 5					2.78		-9.16
					[0.86]		[-2.32]
Anomaly 6						6.81	0.68
						[2.11]	[0.19]
mkt	-1.32	-2.33	-2.77	-3.75	-4.65	-3.76	0.16
	[-0.71]	[-1.23]	[-1.43]	[-2.03]	[-2.50]	[-1.85]	[0.08]
smb	-4.08	-7.18	-7.40	-8.24	-11.64	-12.10	-4.81
	[-1.40]	[-2.45]	[-2.39]	[-2.79]	[-4.16]	[-3.34]	[-1.29]
hml	-27.84	-25.72	-29.04	-26.89	-29.36	-34.64	-29.89
	[-8.15]	[-7.14]	[-8.28]	[-7.44]	[-7.97]	[-9.28]	[-7.33]
rmw	12.80	16.93	15.32	21.69	24.71	20.99	9.61
	[3.08]	[3.99]	[2.87]	[5.56]	[5.32]	[4.25]	[1.61]
cma	8.70	25.27	27.13	23.24	24.17	22.99	10.47
	[1.53]	[4.83]	[5.06]	[4.40]	[4.54]	[4.20]	[1.70]
umd	2.86	1.59	2.93	1.99	2.48	2.39	4.58
	[1.63]	[0.88]	[1.63]	[1.10]	[1.28]	[1.19]	[2.23]
# months	624	620	624	620	620	577	577
$\bar{R}^2(\%)$	32	30	28	29	27	32	38

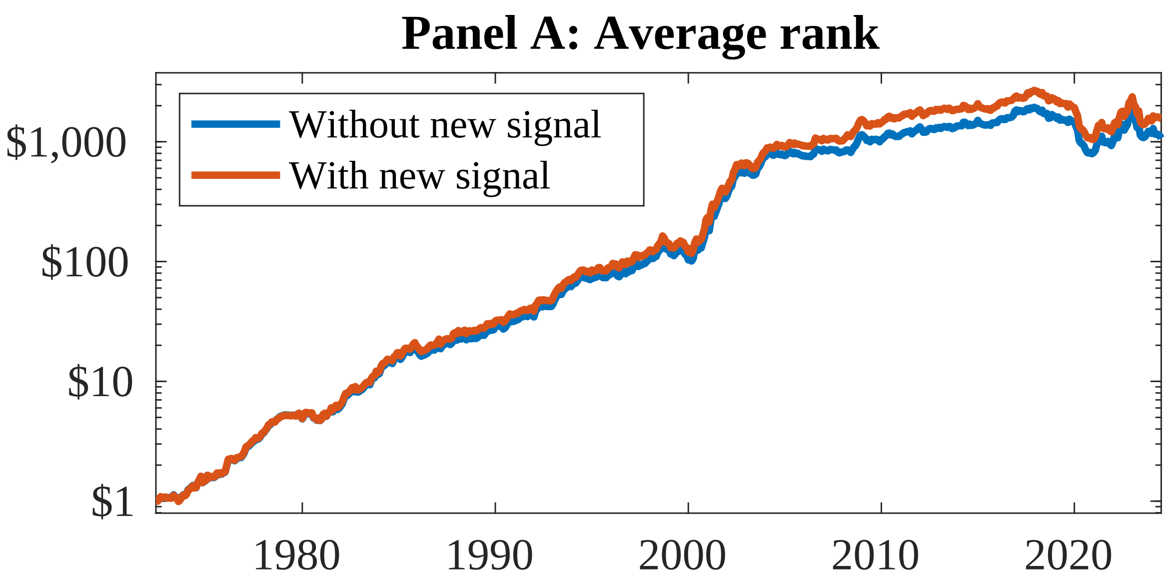


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as NBR. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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